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Home Equity Loan (HMEQ)

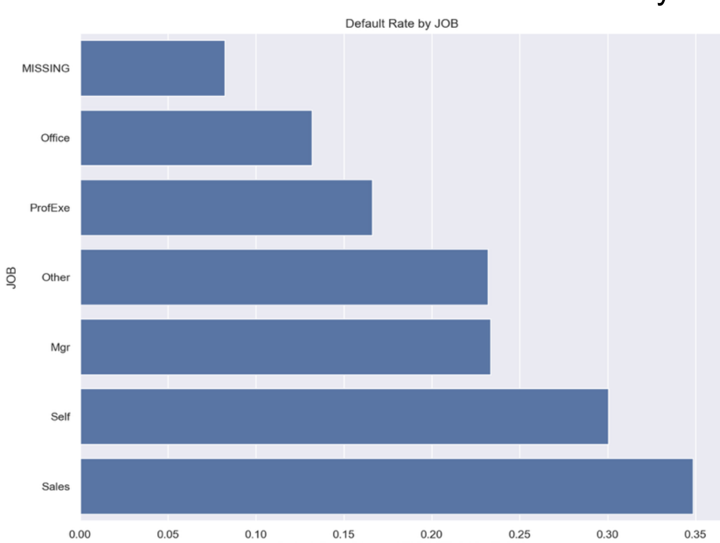
The objective of analyzing the Home Equity ("HMEQ") dataset is to prepare it for predictive modeling to assess loan risk. The dataset that was used contained 5960 rows (loans) and two target variables. One was TARGET_BAD_FLAG, which indicated whether a loan defaulted (1 = default, 0 = repaid), and the other was TARGET_LOSS_AMT, which recorded the loss amount incurred when a loan defaulted (available only for defaulting loans). This dataset included 12 potential predictor variables (10 numeric variables and 2 categorical variables). Whether a loan is defaulted or not can be determined based on the borrower's credit behavior (e.g., delinquencies and high debt-to-income), whereas loss severity is dominated by exposure size, which is led by loan amount (83.7% correlation with loss), and leverage-related variables confirm that different factors explain whether a loan fails versus how costly that failure becomes.

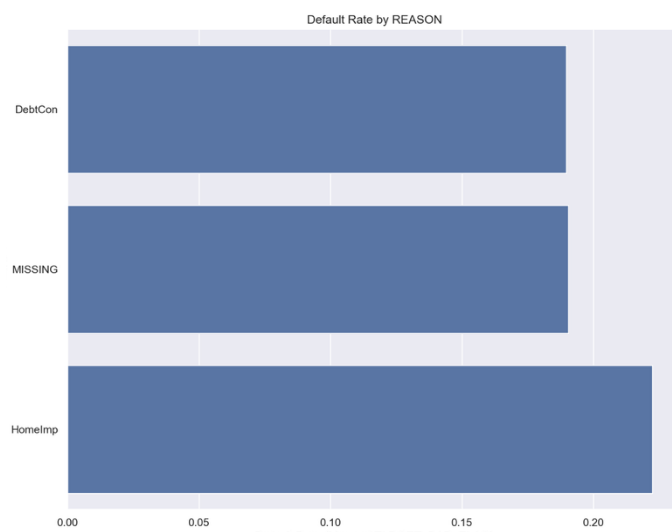
Loan Distribution

Based on the target distribution, the portfolio has a default rate of approximately 19.95% (1,189 defaults out of 5,960 total loans). The loss target is missing for 80.05% of loans because non-defaulting loans do not generate a loss amount.

Categories Driving Default and Loss

The two categorical variables in the dataset are Reason (the purpose for which the loan was taken) and Job (the borrower's job type). Analyzing these helps determine how each category is linked to both loan defaults and loss severity.





From the visuals above, the main reason driving the higher probability for a defaulted loan is Home Improvement (22.25%). As for the type of job with the highest probability of a bad loan, the Borrower is Sales (34.86%). However, once we look at the average loss amount by category, while Home Improvement loans show a higher likelihood of default, Debt Consolidation defaults are associated with a higher loss severity of $16,005 (nearly 2x the

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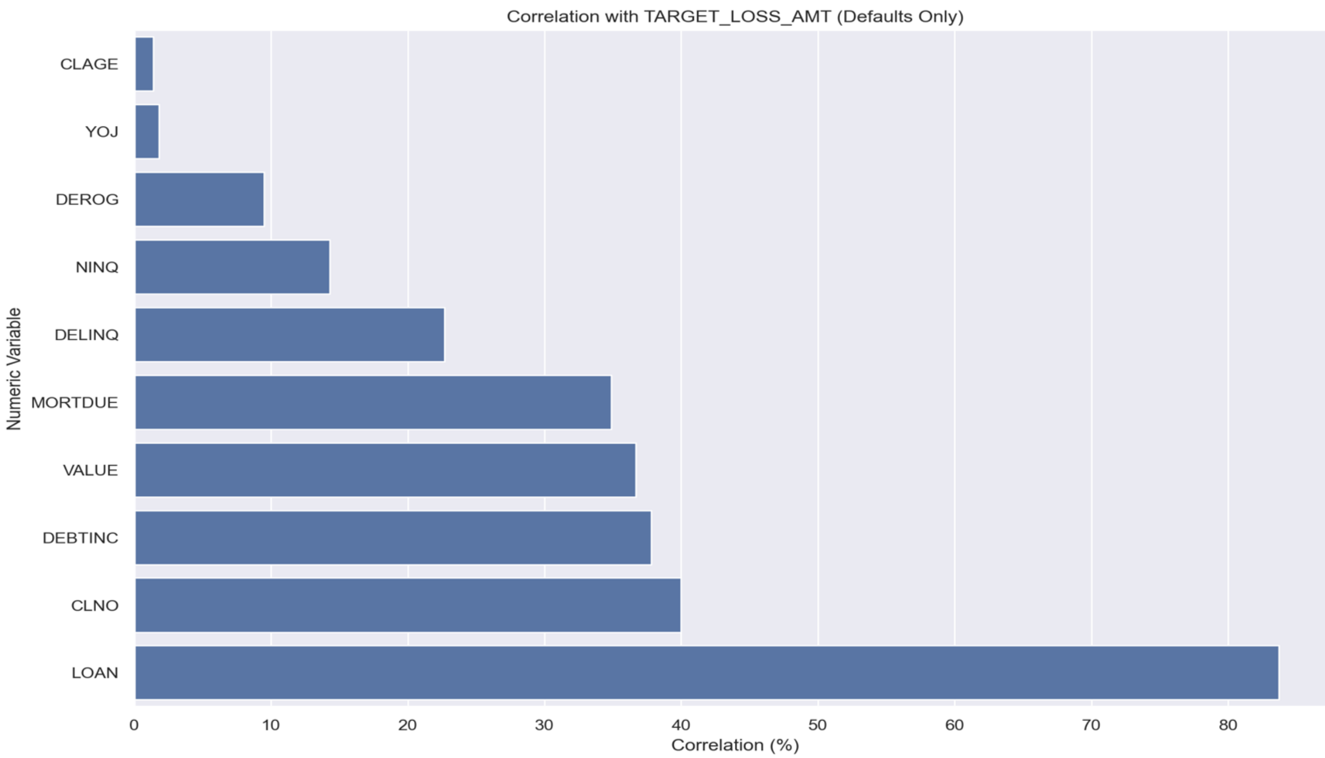
Home Improvement average loss at $8,388). Similar for job category: Self-Employed individuals have the highest average loss severity at $22,232, followed by Sales employees at $16,421. Since Sales jobs are more volatile due to commission-based compensation, there can be more frequent delinquency and missed payments. As for Self-employed jobs, they tend to take larger loans for business related reasons and the predictability of cah flow can be harder in certain cases.



The correlation analysis of numeric variables with loss severity (TARGET_LOSS_AMT), based on defaulted loans only, indicates that after a loan defaults, the amount lost depends mostly on the loan size and the borrower's debt load, rather than on the borrower's past payment behavior. From the chart above, it is clear that the loan amount dominates loss outcomes with a very strong correlation of 83.7%, followed by the number of credit lines, debt-to-income ratio, property value, and remaining mortgage balance, while delinquency and inquiry measures play a comparatively smaller role in determining loss magnitude.